Food Handling and Short-Term Storage at Street-Food Stalls in Gachibowli, Hyderabad: Thermal, Chemical and Solid-Waste Indicators in *Pani Puri* Water

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Abstract

Street food is a defining feature of urban Indian food systems, but the informal conditions under which it is prepared, stored and served make food-safety and environmental outcomes difficult to characterise. We report a cross-disciplinary field and laboratory study of a single, tightly defined item: the spiced water (*pani*) served with *pani puri*, at six vendor stalls distributed across three neighbourhoods of Gachibowli, Hyderabad. At each stall we recorded container material, lid configuration and water temperature over a fixed five-minute window; collected a 30 mL sample for a two-hour open-exposure pH trial; and conducted a disposable-item audit to estimate daily dry-waste mass. Two controlled laboratory trials complemented the field work: a cooling-rate comparison of four container materials (80 cm^3, from 50.0 °C) and a wet-cotton evaporative-cooling trial (35 mL, matched borosilicate beakers). Field water temperatures spanned 20.5-26.5 °C, placing every stall inside the range that supports mesophilic microbial growth. Five of six stalls cooled towards the cool evening ambient; the one stall using a wet-cotton wrap *warmed* by 3.0 °C, consistent with a sub-ambient starting temperature produced by evaporative cooling earlier in the day. In the laboratory, fitted cooling gradients ordered as thin dense ceramic (-2.52), steel (-2.40), thick glass (-1.58) and plastic (-0.26 °C min^{-1}), showing that wall thickness and porosity matter as much as the bulk material; the wet-cotton-wrapped beaker gained only 1.7 °C against 2.4 °C for its uncovered twin (gain ratio 1.41). pH fell in every sample over two hours, by 0.2-0.7 units (mean -0.50), with the largest mean change in Vinayak Nagar (-0.65) and the smallest in Rajiv Gandhi Nagar (-0.30). Estimated dry waste ranged from 179 g d^{-1} (expanded polystyrene) to 836 g d^{-1} (silver-laminated bowls), and tissue use rather than bowl mass dominated the total at the highest-output stalls. No microbiological culture was performed; microbial risk is treated strictly as an inference from measured physicochemical conditions and published growth thresholds. The three indicator families pull in different directions: the materials that cool best are the least hygienic and the most persistent as waste, and the best-performing thermal practice observed (a wet cotton wrap on a steel vessel) is also the cheapest and the lowest-waste. We argue that these trade-offs, not any single indicator, are the useful object of policy attention.

Keywords: street food safety; *pani puri*; evaporative cooling; heat transfer; pH drift; solid waste; informal food vending; Hyderabad; interdisciplinary field study

1 Introduction

Street food vending supplies a large share of daily meals in Indian cities, and it does so through an infrastructure that is almost entirely informal: vessels are chosen by cost and durability rather than by thermal specification, cold storage is usually absent, and waste collection is undifferentiated. The food-safety literature treats temperature control as the single most important lever available to a vendor, yet the vessels and coverings actually in use at street stalls have rarely been characterised quantitatively in the field.

Pani puri is an unusually tractable subject for such a study. The dish pairs a fried hollow shell with a spiced, acidic water (*pani*) that is prepared in bulk at the start of a vending session and then held, uncooked, in an open or loosely covered vessel for several hours. The held liquid is therefore the critical control point: it is the only component that is neither fried nor eaten immediately, it is exposed to ambient air and to repeated ladle contact, and its temperature and chemistry evolve over the vending session. Because the item is standardised across vendors, a single food can be compared across stalls, materials and neighbourhoods without confounding by dish type.

This paper reports a study built around one overarching question:

> *How do food handling and short-term storage conditions in different Gachibowli neighbourhoods influence measurable indicators of food safety and environmental waste?*

The question was decomposed into four measurement strands and one integrative strand, each addressing a sub-question that consumes the output of at least one other strand (Table 1). The design is deliberately constrained. Rather than attempting a broad survey, we fixed the food, the neighbourhoods and the stalls, and measured a small number of indicators well enough to compare them against each other. The contributions are:

1. A field characterisation of container material, covering practice and short-interval thermal behaviour at six stalls, with every observation taken in a matched evening window to hold ambient conditions approximately constant.
2. Two controlled laboratory trials that isolate the mechanisms suggested by the field data: conduction through container walls, and evaporative cooling from a wetted fabric wrap.
3. A paired pH time series under two hours of open exposure, with a distilled-water calibration control and dual-instrument verification.
4. A disposable-item audit converting bowl and tissue counts into daily dry-waste mass, and a material-by-material assessment of what that waste does once discarded.
5. An explicit treatment of the trade-offs between these indicators, including the cases where optimising one indicator demonstrably degrades another.

We also state plainly what this study does not do. No microbiological culture was carried out, so no claim about actual contamination is made anywhere in this paper. Ingredient composition and preparation method were not controlled, because they are vendor-specific and not observable without interfering with the stall. With six stalls, the data support description and

Table 1. Measurement strands, the sub-question each addresses, and the dependency structure of the study design. Every strand consumes the output of at least one other strand.

Strand	Sub-question	Primary output	Consumed by
Thermal	How does container material and insulation affect the rate of heat gain or loss in stored *pani puri* water?	Temperature, rate of change	Chemical, risk inference, integration
Chemical	How does short-term open exposure affect the pH of *pani puri* water across neighbourhoods?	Paired pH, ΔpH	Risk inference, integration
Waste	How do disposal materials at stalls influence daily solid-waste mass and local cleanliness?	Dry-waste mass by component	Integration, synthesis
Integration	How can spatial mapping and basic visualisation reveal patterns across neighbourhoods?	Maps, cross-indicator plots	Synthesis
Risk inference	How do temperature and exposure conditions influence the *potential* for microbial growth?	Condition-to-response mapping	Synthesis

mechanism-testing, not inferential statistics; we report no p-values and treat all neighbourhood differences as descriptive.

2 Materials and methods

2.1 Study area and stall selection

Six vendor stalls were sampled across three adjacent residential neighbourhoods of Gachibowli, Hyderabad: Telecom Nagar, Vinayak Nagar and Rajiv Gandhi Nagar (Figure 1). Two stalls were sampled per neighbourhood. Stalls were selected on three criteria: they sold *pani puri* as a primary item, they were operating at the time of the visit, and the vendor consented to observation and to a small sample being taken. The same six stalls were used by all measurement strands, so that thermal, chemical and waste data refer to the same vendors rather than to separate samples. Stall coordinates were recorded to four decimal places; the two Telecom Nagar stalls sit approximately 900 m southeast of the remaining cluster, and stalls 4-6 lie within roughly 150 m of one another.

2.2 Field thermal observations

At each stall we recorded the storage vessel material, the lid or covering material, and whether the vessel was fully or partially ($\sim$75 %) covered. Water temperature was measured with a digital probe thermometer (resolution 0.1 °C) inserted into the bulk of the stored *pani*, taking an initial reading and a second reading after a fixed interval of five minutes.

The five-minute window was chosen deliberately. A longer observation window at a working stall accumulates confounders that cannot be controlled from outside the stall: the vendor serves customers and removes liquid, bystanders block airflow around the vessel, gusts change the convective boundary condition, and ambient temperature itself drifts. A short,

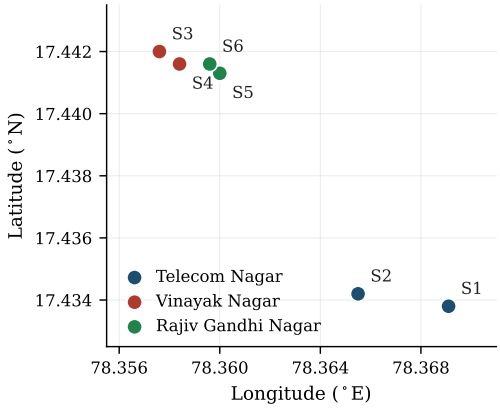


Figure 1. Locations of the six sampled *pani puri* stalls across three Gachibowli neighbourhoods. Stalls 3-6 form a tight northern cluster; the two Telecom Nagar stalls lie to the southeast. All strands sampled the same six stalls.

fixed window holds these approximately constant across stalls at the cost of resolving only a mean rate rather than a full cooling curve. All field observations were made between 17:00 and 19:00 to keep ambient conditions and the vendor's own preparation schedule roughly comparable. Reported rates are mean rates over the window, $\Delta T/5\,\mathrm{min}$, and carry a propagated uncertainty of $\pm 0.2\,^{\circ}\mathrm{C}$ on ΔT.

Sample density was measured for each stall's *pani* by mass and volume, to check whether differences in dissolved and suspended solid content were large enough to alter effective heat capacity and so confound the material comparison.

2.3 Controlled cooling trial

To isolate conduction through the container wall from the many uncontrolled variables of the street setting, four vessels of different materials (plastic, thick glass, steel, and a thin, dense glazed ceramic) were each filled with $80\,\mathrm{cm}^3$ of *pani* prepared at home to a single recipe, eliminating inter-vendor preparation differences. All four samples were brought to a common initial temperature of $50.0\,^{\circ}\mathrm{C}$ in a rectangular water bath (Figure 2), then removed and allowed to cool in still laboratory air with temperature recorded every 30 s for 120 s. A raised initial temperature was used because the driving temperature difference in Newton's law of cooling is proportional to $(T - T_{\mathrm{amb}})$; starting well above ambient makes the material-dependent differences in cooling rate measurable within a short window. Cooling rates are reported as the gradient of a least-squares straight-line fit to temperature against time, converted to $^{\circ}\mathrm{C\,min}^{-1}$.

2.4 Evaporative-cooling trial

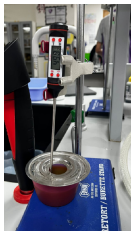
One field stall wrapped its steel vessel in water-soaked cotton fabric, and was the only stall whose water warmed over the observation window. To test the mechanism, two borosilicate beakers of identical geometry were each filled with 35 mL of the same home-prepared *pani* at a common initial temperature of $17.9\,^{\circ}\mathrm{C}$, below laboratory ambient, mirroring the field case. One beaker was left bare; the other was wrapped in water-soaked cotton secured with a band (Figure 3). Both were left undisturbed for 15 min and final temperatures recorded. Because both samples started below ambient, the quantity of interest is the *temperature gain*, and the



(a) Container materials



(b) Water bath equilibration



(c) Probe placement

Figure 2. Laboratory arrangement for the controlled cooling trial. Four vessels of differing material (a) were equilibrated to 50.0 °C in a water bath (b) and then cooled in still air with a clamped digital probe held at a fixed depth (c).



Figure 3. Evaporative-cooling trial: matched borosilicate beakers holding 35 mL of *pani* at 17.9 °C, one bare and one wrapped in water-soaked cotton.

comparison is the ratio of gains.

2.5 pH under short-term open exposure

A 30 mL sample of prepared *pani* was collected from each of the six stalls into a 50 mL plastic beaker (volume uncertainty ±5 mL). A seventh beaker of distilled water, taken as pH 7.0, served as a calibration check (Figure 4). Initial pH was recorded at 09:15 for all samples within a five-minute span. Samples were then left uncovered at room temperature for two hours, and final pH recorded at 11:15, all readings taken within a ten-minute span. Every reading was taken on two independent pH meters and cross-checked to reduce instrumental error; reported values carry an uncertainty of ±0.1 on each reading and ±0.2 on the difference. Volume, beaker type and exposure duration were held constant across samples.

2.6 Disposable-item audit

At each stall, the mass of a single serving bowl and of a single tissue was measured on a digital balance (±0.01 g), and the vendor was asked for the number of bowls and tissues used in a typical day. Daily dry-waste mass was estimated by the count-proxy method:

$$m_{\mathrm{dry}} = n_{\mathrm{bowl}}\,\bar{m}_{\mathrm{bowl}} + n_{\mathrm{tissue}}\,\bar{m}_{\mathrm{tissue}}. \tag{1}$$

Wet waste (chutney residue, soaked shells, discarded *pani*) was not quantified: residue mass per bowl varied too much between servings and between vendors for a count proxy to be

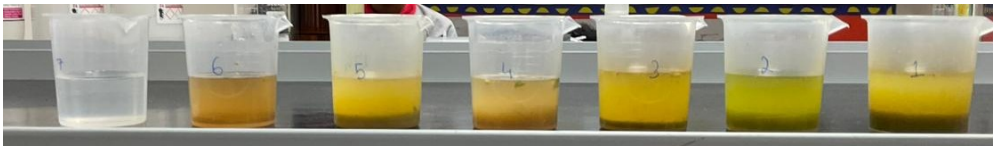


Figure 4. The pH sample set after collection. Beakers 1-6 hold 30 mL of *pani* from the correspondingly numbered stalls; beaker 7 holds distilled water as a calibration check. Visible differences in colour, turbidity and settled solids across samples reflect the uncontrolled variation in vendor recipe and preparation method discussed in Section 5.

defensible, and we prefer to report a narrower quantity well than a broader one badly. Bowl material was recorded as sugarcane bagasse, silver-laminated paper or expanded polystyrene (locally, thermocol). Vendors were also asked informally about waste segregation practice.

2.7 Data integration and visualisation

All strands wrote into a single per-stall record keyed on stall identifier, with fields for coordinates, container and lid class, initial and final temperature, rate of temperature change, initial and final pH, ΔpH, bowl material, item counts and component and total waste mass. Analysis was confined to descriptive statistics and visualisation: paired before/after plots, component-resolved bar charts, bubble maps of each indicator over the stall coordinates, and cross-indicator scatter plots. No modelling, optimisation or machine learning was applied; with $n = 6$, mapping and plotting are used to make differences visible, not to assert relationships. Figures were regenerated from the tabulated values for this paper.

2.8 Microbial-risk inference

Microbiological culture was outside the scope and safe operating limits of the study, so no plate counts or presence/absence tests were performed. Instead, measured temperature and pH conditions were compared against published growth thresholds and the mechanisms by which microbial metabolism alters the chemistry of a food matrix [1, 2]. All statements in this paper about microbial growth are therefore statements about *potential* under the measured conditions, and are reported as such. Where we write that a condition "supports" growth, we mean that published thresholds place it inside a permissive range, not that growth was observed.

3 Results

3.1 Field thermal behaviour

Field observations are summarised in Table 2 and Figure 5. Stored water temperatures spanned 20.5-26.5 °C at the start of the observation window and 21.1-26.1 °C at the end.

Steel dominated as a vessel material (five of six stalls), and every stall covered its *pani*, either partially or fully. Three observations follow directly from the table. First, the distinction between partial ($\sim$75 %) and full covering produced no discernible difference in temperature change, which suggests that over this window the uncovered fraction was not the controlling resistance, although a cover also serves the separate and arguably more important function of keeping foreign matter out. Second, aluminium and steel lids behaved indistinguishably,

Table 2. Field observations of storage configuration and water temperature at six *pani puri* stalls. Temperatures carry an uncertainty of ±0.1 °C, differences ±0.2 °C. Rate is the mean over the five-minute observation window.

				Temperature (°C)			
Stall	**Neighbourhood**	**Vessel**	**Lid / covering**	initial	final	**change**	°C min^{-1}
1	Telecom Nagar	Thick painted ceramic	Thin wood	24.1	23.8	-0.3	-0.06
2	Telecom Nagar	Steel	Steel	24.3	21.1	-3.2	-0.64
3	Vinayak Nagar	Steel	Steel (partial)	24.7	23.8	-0.9	-0.18
4	Vinayak Nagar	Steel + wet cotton	Steel	20.5	23.5	+3.0	+0.60
5	Rajiv Gandhi Nagar	Steel	Aluminium (partial)	26.5	26.1	-0.4	-0.08
6	Rajiv Gandhi Nagar	Steel	Aluminium	24.5	23.6	-0.9	-0.18

Measured sample densities were 1.002, 1.045, 1.027, 1.060, 1.011 and 1.025 g cm^{-3} for Stalls 1-6 respectively. The spread is small enough that differences in effective heat capacity cannot account for the order-of-magnitude range in cooling rate.

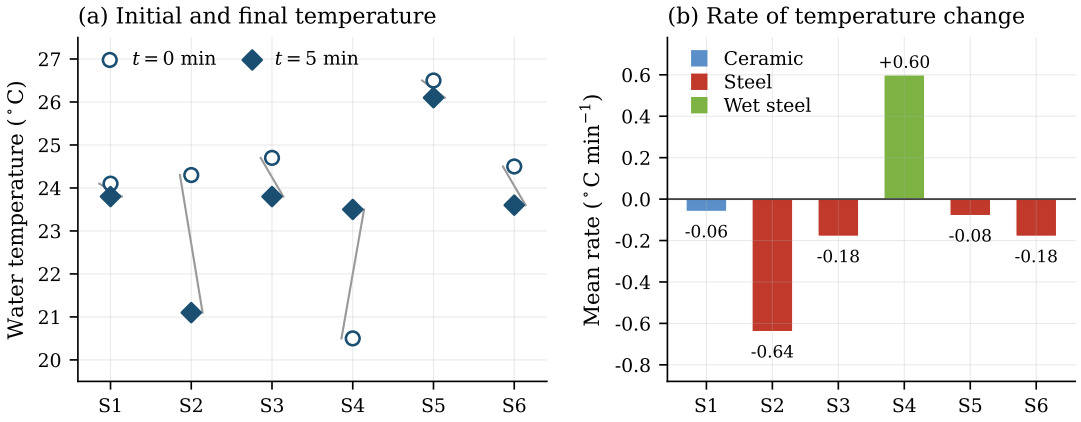


Figure 5. Field thermal behaviour. (a) Paired initial and final water temperature at each stall over the five-minute window. (b) Mean rate of change, coloured by vessel configuration. Stall 4, the only vessel wrapped in wet cotton, is also the only stall whose water warmed.

consistent with the lid being out of direct contact with the liquid and therefore not on the dominant conduction path. Third, the ceramic vessel at Stall 1 showed the smallest magnitude of change among the cooling stalls (-0.06 °C min^{-1}), and the bare steel vessel at Stall 2 the largest (-0.64 °C min^{-1}), an order-of-magnitude spread that tracks the expected ordering of thermal conductivity.

Stall 4 is the outlier and the most informative case: its water *warmed* by 3.0 °C while every other stall cooled. This stall was the only one whose steel vessel was wrapped in water-soaked cotton at the start of the day. Its initial temperature, 20.5 °C, was also the lowest recorded and below the evening ambient. We return to the mechanism in Section 3.3.

3.2 Container material under controlled conditions

The controlled cooling trial (Figure 6a) returned fitted gradients of -0.26 °C min^{-1} for plastic, -1.58 °C min^{-1} for thick glass, -2.40 °C min^{-1} for steel and -2.52 °C min^{-1} for the thin dense

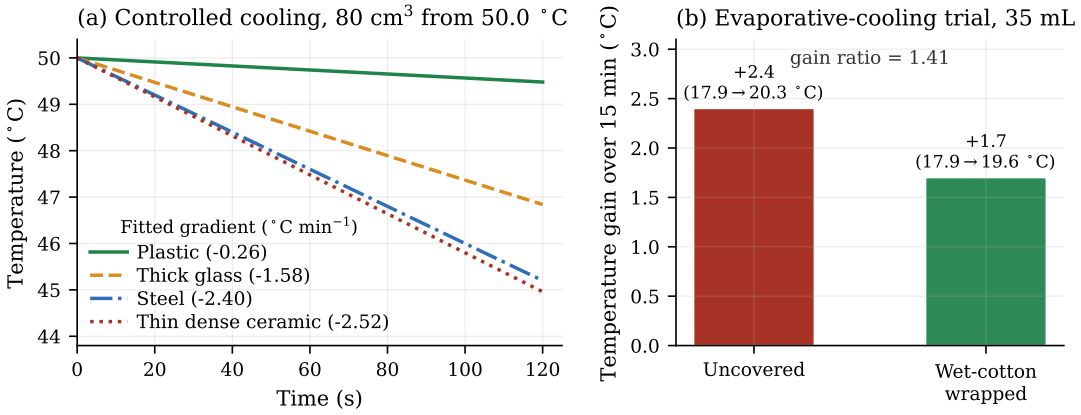


Figure 6. Controlled laboratory trials. (a) Cooling of 80 cm^3 of *pani* from 50.0 °C in four container materials, with fitted gradients; lines show the fits. (b) Temperature gain over 15 min for matched 35 mL samples started at 17.9 °C, bare versus wet-cotton-wrapped.

ceramic. The ordering steel > glass > plastic reproduces the conventional ranking of thermal conductivity for these materials and acts as an internal check on the method.

The ceramic result initially appears to contradict the field observation at Stall 1, where a ceramic vessel retained heat best. The two are reconciled by vessel construction rather than by material class. The laboratory ceramic was a thin, dense, low-porosity glazed piece; the Stall 1 vessel was a thick painted vessel of the kind whose body retains substantial closed porosity. Per unit of wall area, the thin dense piece presents a short conduction path and little trapped air, and so performs poorly as an insulator despite being a ceramic. The practical conclusion is that *wall thickness and porosity are at least as important as bulk material*: thicker, less dense, more porous walls insulate better, and a material label alone does not predict performance.

3.3 Evaporative cooling

In the wet-cotton trial (Figure 6b), both beakers started at 17.9 °C, below laboratory ambient, and both warmed. The bare beaker reached 20.3 °C, a gain of 2.4 °C; the wet-cotton-wrapped beaker reached 19.6 °C, a gain of 1.7 °C. The uncovered beaker therefore gained 1.41 times as much heat as its wrapped twin over fifteen minutes, with geometry, volume, material and initial temperature all matched.

Two conclusions follow. First, the wrap measurably offsets heat transfer into the liquid, which accounts for Stall 4 holding its *pani* below evening ambient. Second, and importantly, the offset is partial: the wrapped sample still warmed. Evaporative cooling did not hold the liquid at temperature, it only slowed its approach to ambient. This matters for the design question in Section 4.5, because it means the wrap and the vessel wall are complementary rather than alternative strategies.

3.4 pH under open exposure

pH fell in every one of the six samples over the two-hour exposure (Table 3, Figure 7). Initial values ranged from 2.5 to 3.7 (the liquid is strongly acidic as prepared, consistent with tamarind and lime bases) and declined by 0.2-0.7 units, with a mean change of -0.50. Mean

Table 3. pH of prepared *pani puri* water before and after two hours of open exposure at room temperature. Readings ±0.1, differences ±0.2.

Sample	Neighbourhood	Initial pH	Final pH	ΔpH	Neighbourhood mean
1	Telecom Nagar	3.2	2.5	-0.7	-0.55
2	Telecom Nagar	2.6	2.2	-0.4	
3	Vinayak Nagar	2.8	2.1	-0.7	-0.65
4	Vinayak Nagar	2.7	2.1	-0.6	
5	Rajiv Gandhi Nagar	3.7	3.3	-0.4	-0.30
6	Rajiv Gandhi Nagar	2.5	2.3	-0.2	
All samples				**-0.50**	

Initial readings taken at 09:15 within a five-minute span; final readings at 11:15 within a ten-minute span. A seventh beaker of distilled water (pH 7.0) served as calibration control.

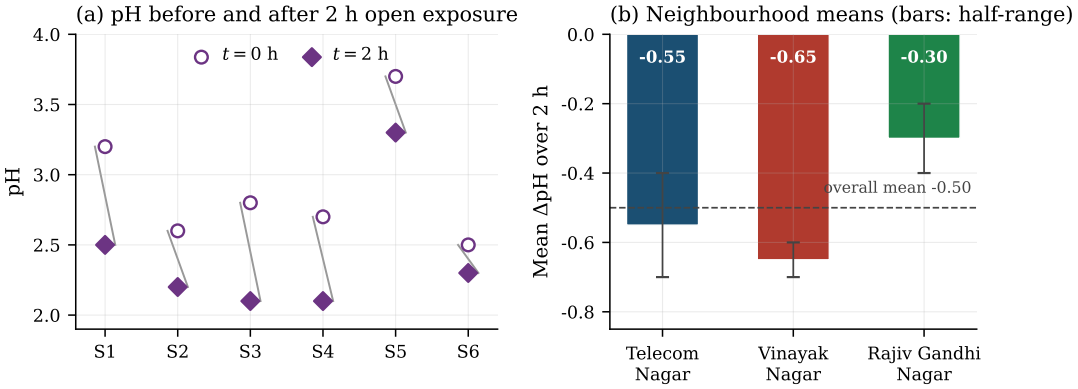


Figure 7. pH under two hours of open exposure. (a) Paired initial and final pH for all six samples; every sample acidified. (b) Neighbourhood mean ΔpH, with bars showing the half-range of the two samples contributing to each mean.

changes by neighbourhood were -0.55 (Telecom Nagar), -0.65 (Vinayak Nagar) and -0.30 (Rajiv Gandhi Nagar). The Rajiv Gandhi Nagar samples were thus the most chemically stable over the window, and the Vinayak Nagar samples the least, a difference of 0.35 pH units between neighbourhood means.

The distilled-water control read 7.0 as expected, and dual-meter cross-checking produced no discrepancies beyond the stated ±0.1 resolution. We note that the direction of change was uniform across all six samples while the magnitude varied by a factor of three; the uniformity of direction is the more robust finding, since magnitude is sensitive to the uncontrolled recipe differences visible in Figure 4.

3.5 Daily dry-waste mass

The disposable-item audit (Table 4, Figure 8) returned estimated daily dry-waste masses between 178.75 g (Stall 5) and 836 g (Stall 4), a factor of 4.7. Four of six stalls used silver-laminated bowls; one used sugarcane bagasse and one used expanded polystyrene.

Table 4. Disposable-item audit and estimated daily dry-waste mass by component. Item masses ±0.01 g. Component masses computed by Equation (1).

		Unit mass (g)		Items per day		Dry waste ($g\,d^{-1}$)		
Stall	**Bowl material**	bowl	tissue	bowls	tissues	bowls	tissues	total
1	Sugarcane bagasse	2.09	1.06	200	280	418.0	296.80	714.80
2	Silver-laminated	1.79	0.83	100	750	179.0	622.50	801.50
3	Silver-laminated	1.69	0.75	200	100	338.0	75.00	413.00
4	Silver-laminated	1.76	0.88	200	550	352.0	484.00	836.00
5	Polystyrene (EPS)	0.50	0.73	175	125	87.5	91.25	178.75
6	Silver-laminated	1.70	0.69	100	200	170.0	138.00	308.00

Tissue counts at Stalls 2 and 4 are midpoints of vendor-reported ranges (700-800 and 500-600 respectively). Wet waste was not quantified owing to inconsistency between servings.

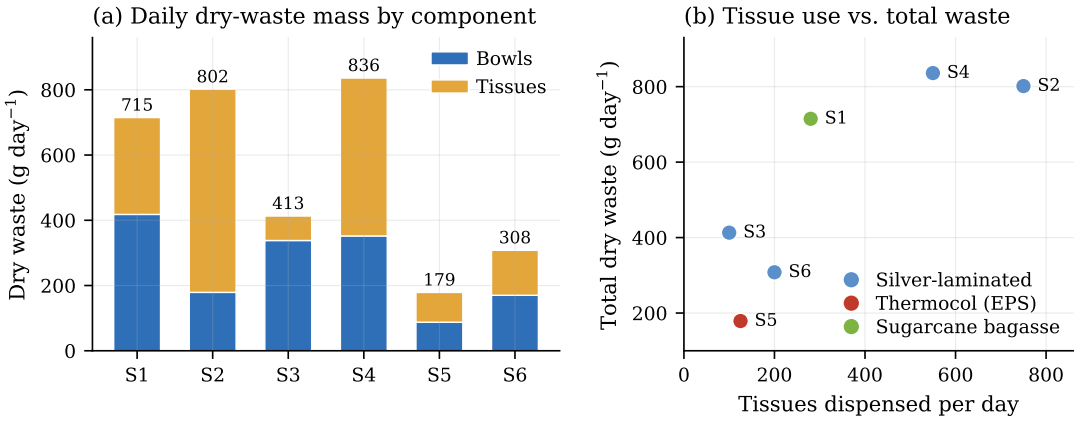


Figure 8. Estimated daily dry waste. (a) Component-resolved mass by stall; tissue mass exceeds bowl mass at the two highest-output stalls. (b) Total dry waste against tissues dispensed per day, coloured by bowl material: tissue use, not bowl material, separates the high- and low-output stalls.

The component breakdown is the key result. Bowl mass alone does not explain the spread: Stalls 2 and 6 used the same silver-laminated material and the same bowl count scale, yet generated 801.5 g and 308 g respectively. The difference is tissues: 750 per day against 200. At the two highest-output stalls, tissue mass exceeded bowl mass (622.5 g versus 179 g at Stall 2; 484 g versus 352 g at Stall 4). Tissues are supplied to improve serving hygiene, and in doing so they more than double the daily dry-waste load. This is the first of the trade-offs we take up in Section 4.5.

Stall 5's polystyrene bowls produced the lowest total mass by a wide margin, at 0.50 g per bowl against 1.69-2.09 g for the alternatives. Mass, however, is a poor proxy for environmental burden in this case, as Section 4.4 develops.

Grouping the same values by material rather than by stall (Figure 9) isolates what the material itself contributes. Neither grouping is clean. The four steel vessels span the full range of pH change in the sample, from -0.2 to -0.7, so vessel material does not sort the chemical result; and the four silver-laminated stalls span 308 to 836 $g\,d^{-1}$, a factor of 2.7, so

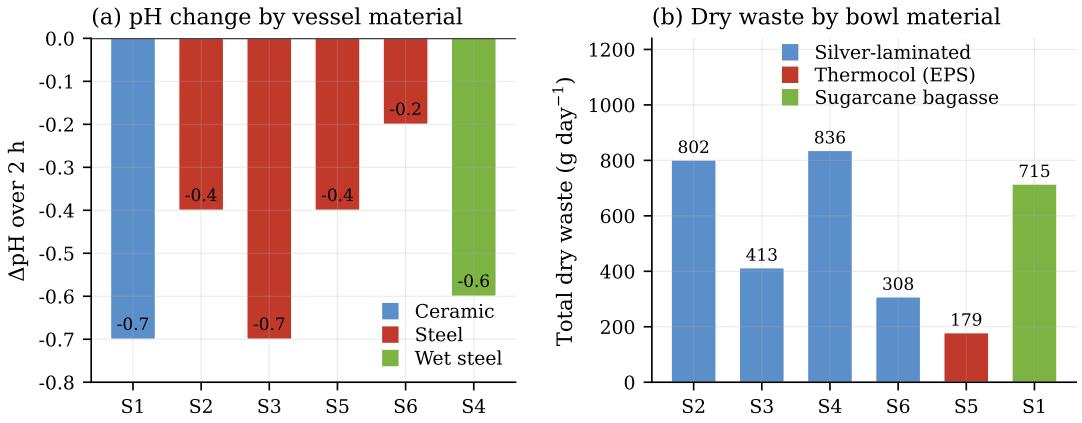


Figure 9. The same indicators grouped by material rather than by stall. (a) pH change over two hours, by vessel material class. (b) Total daily dry waste, by bowl material. Within-material spread is comparable to between-material spread in both panels.

bowl material does not sort the waste result either. Within-material variation is comparable to between-material variation for both indicators, which is the quantitative form of the point made above: at this sample size, what a vendor does matters more than what the vendor's equipment is made of.

3.6 Spatial and cross-indicator patterns

Mapping the three indicator families onto stall coordinates (Figure 10) shows that they do not co-locate. The largest pH changes occur at Stall 1 (Telecom Nagar) and Stall 3 (Vinayak Nagar), which are at opposite ends of the sampled area. The largest-magnitude temperature rates occur at Stalls 2 and 4, again in different neighbourhoods. Waste output is high at Stalls 1, 2 and 4 and low at Stall 5, with the two highest and the lowest all in different neighbourhoods. Neighbourhood, in other words, does not organise the indicators; stall-level practice does.

Cross-indicator plots (Figure 11) make the same point in a different projection. No monotone relationship appears between thermal behaviour and initial acidity, nor between the magnitude of pH change and daily waste output. Stall 5 is instructive: it has the highest initial pH (3.7), the smallest pH change (-0.4), near-zero temperature change and the lowest waste mass, and yet uses the material with the worst end-of-life profile. A stall can score well on every indicator we measured directly while still representing the largest environmental liability in the sample, which is an argument for measuring more than one thing.

4 Discussion

4.1 Heat-transfer mechanisms in street storage vessels

The two thermal results (a tenfold spread in field cooling rate and a material ordering in the laboratory that is modified by wall construction) are both explicable from the conduction mechanism available to each material class.

In metals, thermal energy is carried both by lattice vibration and by the motion of delocalised electrons through the ion lattice; the electronic contribution dominates and makes steel

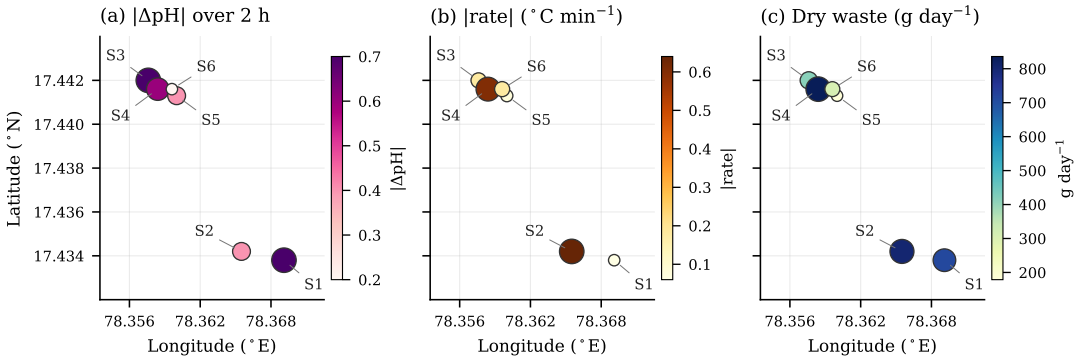


Figure 10. Indicator values mapped onto stall coordinates. (a) Magnitude of pH change over two hours. (b) Magnitude of the rate of temperature change. (c) Estimated daily dry waste. Marker size and colour both encode the value. The three indicator families do not co-locate across neighbourhoods.

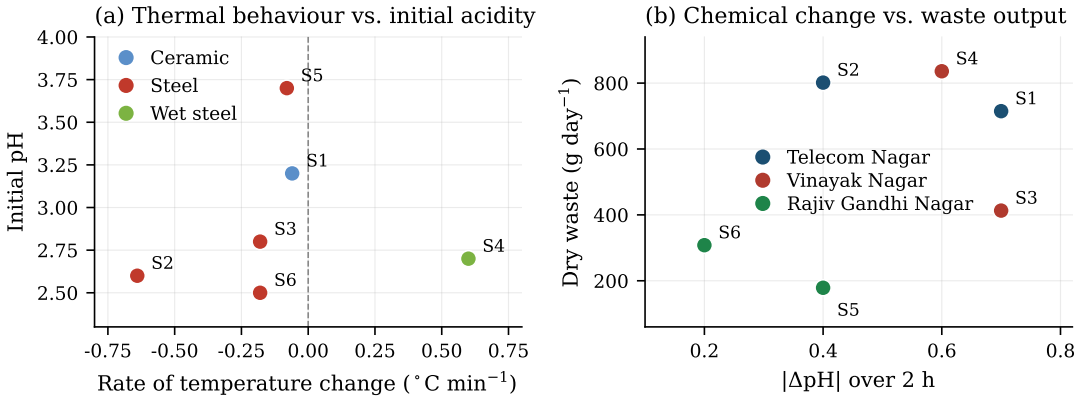


Figure 11. Cross-indicator relationships. (a) Rate of temperature change against initial pH, by vessel configuration. (b) Magnitude of pH change against daily dry-waste mass, by neighbourhood. Neither projection shows a monotone relationship at $n = 6$.

and aluminium efficient conductors. In ionic and covalent-network solids such as ceramics there are no delocalised electrons, and conduction proceeds only through the vibration and collision of strongly bound particles [5, 6]. Strong bonding suppresses vibrational amplitude, which lowers conductivity; but conduction also rises with particle packing density, which is why a thin, dense ceramic can conduct comparatively well while a thick, porous one insulates. Closed porosity is decisive because the trapped air in the pores is itself a poor conductor and the pores interrupt the conduction path: the vessel wall behaves as a composite, not as its nominal material. This accounts for the apparent conflict between the field and laboratory ceramic results without requiring either to be discarded.

The field heat flow runs from liquid to vessel wall, through the wall, and then to the surrounding air and the ground. Because the lid sits above the liquid surface and is not in direct contact with it, it is not on this dominant path, which is consistent with our finding that lid material made no measurable difference. Newton's law of cooling, with rate proportional to $(T - T_{\text{amb}})$, also justifies the two design decisions that make the comparisons interpretable: a common initial temperature in the laboratory trial, and a fixed evening window in the field

so that T_{amb} was approximately constant across stalls.

Evaporative cooling operates by a different route. The most energetic molecules at the surface of the wetted fabric escape to the vapour phase, carrying away the latent heat of vaporisation and lowering the mean kinetic energy, and hence the temperature, of what remains, with no heat removed to a colder body. The heat for that phase change is drawn preferentially from the vessel and its contents rather than from the surrounding air, because the solid-to-solid conduction path (vessel wall to damp fabric) is far more efficient than the air-to-solid path. This is the operating principle of the unglazed *matka* and of desert coolers, and it is what allowed Stall 4 to hold its *pani* below evening ambient. Our trial confirms the offset but also bounds it: at 1.41 times the heat gain in the bare control, the wrap slows the approach to ambient without arresting it.

4.2 Interpreting the acidification

Every sample acidified over two hours. Several mechanisms are consistent with the direction of change and we cannot separate them with the measurements taken:

1. **Fermentative metabolism.** Lactic acid bacteria and related organisms ferment carbohydrates to organic acids, and are well characterised as agents of acidification in food matrices [2]. Prepared *pani* contains accessible carbohydrate from its spice and flour components.
2. **Hydrolysis of carbohydrate and starch** liberating acidic products over time.
3. **Dissolution of atmospheric carbon dioxide** in the open beakers, forming carbonic acid, and the products of anaerobic respiration within the liquid.
4. **Limited buffering capacity.** A weakly buffered matrix will register a larger pH shift for the same quantity of acid produced, which is a plausible contributor to the threefold range in ΔpH magnitude across stalls.

Because microbial presence was not measured, the fermentative explanation is a hypothesis consistent with the data rather than a conclusion, and the observed pH drop is reported here as a chemical result with a microbiological interpretation attached, not as evidence of contamination. We note the asymmetry in confidence between the two features of the data: the uniform *direction* of change across six independently prepared samples is hard to explain by measurement artefact, whereas the *magnitude* ordering is confounded by recipe differences we did not control.

4.3 Inferred microbial risk

Mapping the measured conditions onto published growth thresholds gives the picture in Table 5. Every stall held its *pani* between 20.5 °C and 26.5 °C, which lies inside the range in which mesophilic organisms grow and well inside the 5-60 °C interval that food-safety guidance identifies as permissive and advises against holding food within [1, 4]. Cooling over the observation window, at most 3.2 °C, narrowed but did not escape this range. No stall came close to refrigeration temperature, and none could have: none had refrigeration.

Working against this, the matrix is strongly acidic. At pH 2.5-3.7, and falling, the *pani* sits below the growth minima of most bacterial foodborne pathogens, and acidification over the

Table 5. Measured conditions compared against published growth thresholds. Entries describe *potential* only; no microbiological measurement was made in this study.

Condition	Measured range	Published response
Holding temperature	20.5-26.5 °C	Within the mesophilic growth range and inside the 5-60 °C interval that guidance identifies as permissive for growth [4]; growth rate rises with temperature across this interval [1].
Temperature change over window	-3.2 to +3.0 °C	Insufficient to leave the permissive range in either direction; no stall approached refrigeration.
pH as prepared	2.5-3.7	Below the growth minima of most bacterial foodborne pathogens; permissive for acid-tolerant fermentative organisms, moulds and yeasts.
pH drift over 2 h	-0.2 to -0.7	Consistent with organic-acid production by fermentative metabolism [2]; moves the matrix further from bacterial pathogen minima.
Exposure time	2 h open, longer in practice	Holding time is the multiplier on all of the above; a permissive temperature matters more the longer the matrix is held.

holding period moves it further from them rather than closer. Acid tolerance is not uniform, however: acid-tolerant and fermentative organisms, including the lactic acid bacteria whose metabolism is the most parsimonious explanation for the pH drop itself, remain viable in this range, as do acid-tolerant moulds and yeasts. The reasonable inference is therefore not that the *pani* is a high-risk matrix for pathogen proliferation, but that it is a matrix in which temperature is permissive while pH is restrictive, and in which the organisms most likely to be active are spoilage and fermentative rather than pathogenic. Risk at the stall is correspondingly more likely to enter through routes we did not measure (water quality, hand and utensil contact, the shells and fillings) than through growth in the *pani* itself.

4.4 What the waste materials do after disposal

Mass alone ranks the three bowl materials in an order that inverts once end-of-life behaviour is considered, and the inversion depends almost entirely on whether dry and wet waste are separated.

Sugarcane bagasse (Stall 1, 714.8 $\mathrm{g\,d^{-1}}$) has a natural fibre structure that absorbs food moisture without releasing harmful compounds into the food. Kept dry, it composts fully in roughly 30-90 days into a usable soil amendment. Mixed with wet waste, the picture changes: chutney and *pani* residues soak into the fibre and block the air and microbial access that rapid decomposition requires, leaving the material soggy for 6-12 months, generating odour, attracting flies and rodents, and breaking down only 30-40% of the way [7, 8]. It was also the most expensive option, and Stall 1 was in the most commercially affluent part of the sampled area; it is a willingness-to-pay pattern worth noting with a single observation's worth of caution.

Silver-laminated bowls (four stalls, 308-836 $\mathrm{g\,d^{-1}}$) are a three-layer composite: paper for

structure, an aluminium foil layer to stop oil and water seepage, and a thin polymer film. The laminate performs its food-contact job well, which is precisely why it is popular. The same structure defeats recycling, because the layers must be separated before either the paper or the metal can be recovered. Clean and dry, 40-50% is recoverable; contaminated with oily residue, the layers are effectively glued shut and recovery falls to 0-10%, with the material fouling sorting machinery and, when it reaches open drains, contributing to blockages and reduced street cleanliness [8, 9]. On a persistence basis, these bowls are arguably a worse outcome than the polystyrene at Stall 5 despite looking cleaner at the point of service.

Expanded polystyrene (Stall 5, 178.75 $\mathrm{g\,d^{-1}}$) is the lightest and so the best-performing material on a mass metric, and the worst on almost every other. It is not biodegradable and is not recyclable at small scale; kept dry, perhaps 20-25% can be reprocessed by melting, but moisture from mixed waste makes it crumble into fragments that disperse as microplastics and persist in soil and water on century timescales. Routes to chemical upcycling of polystyrene waste exist and are an active research area, but they presuppose collection and separation infrastructure that does not reach an individual street vendor [3]. Polystyrene can also release styrene on contact with hot or oily food, giving it a direct food-safety dimension that the waste-mass figure conceals entirely.

The common thread is segregation. For all three materials, the gap between the best and worst end-of-life outcome is set less by the material than by whether it is kept dry. Vendors we spoke to informally expressed willingness to separate dry and wet waste, but reported that municipal collection takes everything mixed in a single morning round. The constraint is therefore not vendor intent but collection design: a municipal system that collects mixed waste can override every material-level sustainability choice an individual vendor makes, which is an argument for directing intervention at the collection step rather than at vendor behaviour alone.

4.5 The trade-off structure

Read together, the strands describe a set of trade-offs rather than a single optimum. Three are sharp enough to state directly.

Thermal performance against hygiene. The physics points to plastic as the best insulator available cheaply: it had by far the lowest cooling gradient and is inexpensive and durable enough for street use, where ceramic and glass are too fragile. But plastic is the worst of the candidate materials on hygiene. Its surface develops micro-grooves under repeated use and cleaning that are larger, rougher and more numerous than those on well-finished steel, providing refuge for microbial films that survive routine washing, and it can affect taste. A vendor who optimises for cold *pani* by switching to plastic buys a colder and more pleasant product in the short run at the cost of a vessel that is harder to keep clean over a season, and a waste problem at the end of it.

Hygiene against waste. The tissue result is the clearest case in our data. Tissues are provided for hygiene at the point of service, and they are the single largest contributor to daily dry-waste mass at the highest-output stalls, exceeding bowl mass at Stalls 2 and 4. The same logic applies to the silver laminate, which is used because it prevents oil leakage (a real food-contact benefit) and is near-unrecyclable for exactly the structural reason that makes it work.

Mass against persistence. Polystyrene minimises waste mass and maximises waste persistence. Any policy instrument that targets tonnage alone would rank Stall 5 as the best performer in the sample.

The wet cotton wrap is the one intervention in our data that does not sit on a trade-off. It measurably reduces heat gain, it requires no change of vessel and so does not import plastic's hygiene penalty, it can be made from scrap fabric or discarded clothing at effectively zero cost, and it adds nothing to the waste stream. Cotton is a good choice of fabric specifically because it holds the most water per unit mass among cheap, available options. Since the wrap only partially offsets heat gain, it complements rather than replaces the vessel's own insulation: the configuration suggested by our results is a thick-walled, porous, non-plastic vessel under a damp cotton wrap, which is close to what Stall 4 had arrived at by practice.

4.6 Relation to the Sustainable Development Goals

The findings bear on three goals. For **SDG 3** (good health and well-being), the measured holding temperatures place every stall inside a growth-permissive range throughout its vending session, while the strongly acidic matrix restricts bacterial pathogens; this locates the actionable risk in handling practice and in the non-*pani* components rather than in the held liquid. For **SDG 11** (sustainable cities and communities), unsegregated laminate and polystyrene waste contributes to drain blockage and reduced street cleanliness; a stall generating $836\,\mathrm{g\,d^{-1}}$ of mixed dry waste is a non-trivial local load where collection is undifferentiated. For **SDG 12** (responsible consumption and production), the dominant lever in our data is reduction and separation rather than substitution: proper dry/wet segregation moves recyclability from 0-10% to 40-50% for the most-used material in the sample, a larger gain than switching material while continuing to mix.

Three interventions follow from the data rather than from general principle: subsidise or otherwise close the cost gap on biodegradable alternatives, which Stall 1's experience suggests is the binding constraint on adoption; provide separate dry and wet collection at vending clusters, since vendor willingness already exists and collection is the limiting step; and target tissue consumption specifically, as the largest single component of dry waste at the stalls that generate the most.

5 Limitations

The constraints on this study are substantial and bear directly on how far the results generalise.

Sample size and inference. Six stalls across three neighbourhoods support description and mechanism-testing, not statistical inference. Reported neighbourhood means rest on two observations each; we give half-ranges rather than standard errors for that reason, and treat all between-neighbourhood differences as descriptive.

Uncontrolled preparation. Ingredient composition, spice loading, water source and preparation time were vendor-specific and not controlled. This is visible directly in Figure 4, where colour, turbidity and settled-solid content differ markedly between samples, and it is the most likely source of the threefold range in ΔpH magnitude. The laboratory trials used a single home recipe precisely to remove this variable, at the cost of no longer being a street sample.

Field confounders. Vessel geometry was not uniform across stalls, ambient temperature

was held approximately rather than exactly constant, and stirring was not standardised. The Stall 2 result is a concrete instance: that vendor stirred the vessel repeatedly to prevent sedimentation of a relatively solid-rich *pani*, which would promote convective mixing and heat loss, and may contribute to the largest cooling rate in the sample. Sample densities spanned only 1.002-1.060 $\mathrm{g\,cm^{-3}}$, so differences in heat capacity cannot account for the observed spread; stirring, geometry and surface-area-to-volume ratio, however, were not separated from material.

Observation window. Five minutes resolves a mean rate, not a cooling curve, and cannot capture behaviour over a full vending session. The direction of the field result at Stall 4 depends on an inference about conditions earlier in the day that we did not observe directly.

Self-reported counts. Daily bowl and tissue counts came from vendors, and two were reported as ranges, for which midpoints were used. These are estimates of typical days, not audited counts, and no wet waste was quantified at all.

No microbiology. The single largest limitation. No culture, plate count or presence/absence testing was performed, so every statement about microbial growth in this paper is an inference from physicochemical conditions and published thresholds. The fermentative explanation for the pH drop remains a hypothesis.

Laboratory-to-field transfer. The controlled trials used different volumes (80 $\mathrm{cm^3}$ and 35 mL) and in one case a much higher initial temperature (50.0 °C) than any field condition. They were designed to isolate mechanisms, and the mechanisms transfer; the absolute rates do not.

6 Conclusions

We characterised food handling and short-term storage at six *pani puri* stalls across three Gachibowli neighbourhoods, measuring thermal behaviour, pH drift under open exposure and daily dry-waste mass on the same vendors, and testing the suggested mechanisms in two controlled trials.

Stored *pani* was held between 20.5 °C and 26.5 °C at every stall, inside the range published guidance identifies as permissive for microbial growth, and the short-interval change at any stall was too small to leave that range. Cooling rate varied tenfold across stalls and tracked vessel construction rather than material label: the controlled trial ordered steel, glass and plastic as expected, but a thin dense ceramic cooled fastest of all, establishing that wall thickness and porosity matter as much as bulk material. The one stall wrapping its vessel in water-soaked cotton was the only one whose water warmed over the window, and our matched-beaker trial reproduced the mechanism, with the bare control gaining 1.41 times as much heat. pH fell in all six samples, by 0.2-0.7 units over two hours, uniformly in direction and variably in magnitude. Estimated dry waste ranged from 179 to 836 $\mathrm{g\,d^{-1}}$, and was driven by tissue consumption rather than by bowl material at the highest-output stalls.

The result we would emphasise is not any single indicator but the structure connecting them. The material that insulates best is the least hygienic; the practice that improves serving hygiene dominates the waste stream; the material that minimises waste mass maximises waste persistence; and the end-of-life outcome for every material is governed less by the material than by whether a municipal system that collects mixed waste permits it to be kept dry. The

one practice we found that improves an indicator without degrading another (a damp cotton wrap on a non-plastic vessel, assembled from scrap at no cost) was already in use at one stall, arrived at by vendor practice rather than by specification. That seems to us the more useful finding: the available improvements in this system are cheap, and the binding constraints are in collection design and material cost rather than in vendor intent.

Two extensions would address the largest gaps. Culture-based microbiological measurement, under appropriate supervision, would convert the risk inference of Section 2.8 into a measurement. Longitudinal observation across a full vending session, rather than a five-minute window, would resolve the cooling curve that our design can only sample.

Author contributions

A.R. designed and conducted the thermal field measurements and both controlled laboratory trials, and led the heat-transfer analysis. **N.S.K.** designed and conducted the pH measurements, including calibration and dual-meter verification, and led the chemical interpretation. **D.C.** designed and conducted the disposable-item audit and vendor interviews, and led the waste and end-of-life material analysis. **S.G.** led the microbial-risk inference, the condition-to-response mapping of Table 5, and the materials and hygiene trade-off analysis. **A.S.** built the integrated per-stall dataset and produced the spatial maps and cross-indicator visualisations. All authors coordinated stall selection so that every strand sampled the same six vendors, contributed to the synthesis, and reviewed the manuscript.

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Data availability

All data analysed in this study are reported in full in Tables 2-4 and in the figure captions. No data were excluded.

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